



NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M. Tech (MMD) (CBCS) Degree Examinations

Summer Semester July 2024

22MMD223 – FRACTURE MECHANICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Explain how the Boston molasses tank disaster and liberty ship failures triggered the fracture study in materials.	10	L*2	1	1
	b)	List the various NDT methods used in fracture mechanics. With neat sketch, explain any two NDT method of testing.	10	L2	2	1
2.	a)	A sheet of glass 2m long measuring 200mm X 2mm contains a central slit parallel to the 200 mm side. The sheet is restrained at one end and loaded in tension with a mass of 500 kg. What is the maximum allowable length of slit before fracture occurs? Assume plane stress condition and the following material property values: E = 60GPa, surface energy is 0.5 J/m ² .	5	L3	1	2
	b)	Differentiate clearly between cleavage fracture and ductile fracture.	5	L2	1	1
	c)	Consider an infinite periodic array of equally spaced cracks along the z-axis in an infinite plate subjected to equal uniform stresses σ along the z- and y-axes at infinity. Verify that the Westergaard function is				
		$Z_I = \frac{\sigma \sin\left(\frac{\pi z}{W}\right)}{\left[\sin^2\left(\frac{\pi z}{W}\right) - \sin^2\left(\frac{\pi a}{W}\right)\right]^{1/2}}$				
		Then show that the stress intensity factor				
		is given by				
		$K_I = \sigma(\pi a)^{1/2} \left(\frac{W}{\pi a} \tan \frac{\pi a}{W}\right)^{1/2}$	10	L3	2	2
3.	a)	Explain the sources of micro and macro cracks, in materials.	4	L2	1	1
	b)	What are the classifications of the different modes of loading at the crack-tip? Explain with neat sketches	6	L2	1	1
	c)	Verify that the Westergaard function for an infinite plate with a crack of length				

2a subjected to a pair of forces at $x = b$ is $Z_1 = \frac{P}{\pi(z-b)} \left(\frac{a^2 - b^2}{z^2 - a^2} \right)^{1/2}$. Then show that the stress intensity factor of the tip $x = a$ is given by

$K_I = \frac{P}{(\pi a)^{1/2}} \left(\frac{a+b}{a-b} \right)^{1/2}$. Use these results to show that for an additional pair of forces at $x = -b$ the Westergaard function is

$$Z_1 = \frac{2Pz}{\pi(z^2 - b^2)} \left(\frac{a^2 - b^2}{z^2 - a^2} \right)^{1/2}$$

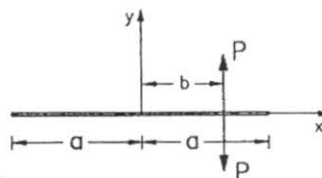


Figure .3(c).

10 L3 2 2

P.T.O.

Unit – II

4. a) Derive an expression for the size of the plastic zone and the correction according to Irwin's approach. State also the significance of it. 10 L2 3 1
- b) Determine the J-Integral for a double cantilever beam specimen, if each cantilever is pulled by a distributed load P , as shown in Figure 4 (b).

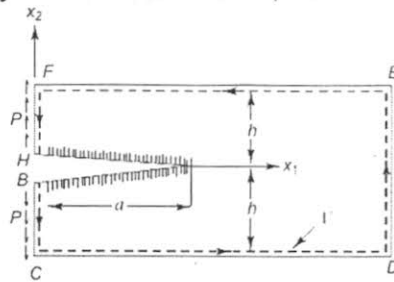


Figure. 4 (b).

5. a) A large plate of 5mm thickness made of medium carbon steel $\sigma_{ys}=350\text{MPa}$ with a through thickness center crack of $2a=40\text{mm}$ length subjected to a stress of 150MPa . Find the effective crack length using Irwin's plastic zone correction. 6 L3 3 2
- b) Discuss the effect of thickness on fracture toughness with neat graphs. 4 L2 3 1
- c) Derive the relationship between CTOD, K_I and G_I . 10 L3 4 1
6. a) Using Dugdale's strip yield model and show that $\rho = (a\pi^2\sigma^2)/(8\sigma_{ys}^2)$. 10 L3 3 2
- b) With neat sketch explain any two techniques used for arresting crack propagation. 10 L2 4 1

Unit – III

7. a) A K_{IC} test is performed on a material with $\sigma_{YS}=690\text{MPa}$. The toughness of this material is expected to lie between 40 and $60\text{MPa}\sqrt{\text{m}}$. Design an experiment to measure K_{IC} (in accordance with ASTM E399) in this material using a compact specimen. Specify the following quantities: (a) specimen dimensions, (b) pre-cracking loads, and (c) required load capacity of the test machine. 10 L4 5 2
- b) A CTOD test was performed on a three-point bend specimen with $B = W = 25.4\text{ mm}$. The crack depth 'a' was 12.3 mm . Examination of the fracture surface revealed that the specimen failed by cleavage with no prior stable crack growth. Compute the critical CTOD in this test. $V_p=1.05\text{ mm}$; $P_{critical}=24.6\text{ kN}$; $E=207\text{ GPa}$; $\sigma_{YS}=400\text{MPa}$; $\nu=0.3$. 10 L3 5 2
8. a) Explain the procedure for experimental determination of CTOD. 10 L2 5 1
- b) A three-point bend specimen with $S = 25\text{ cm}$, $W = 6\text{ cm}$, $a = 3\text{ cm}$, and $B=3\text{ cm}$ is used to determine the critical crack opening displacement δ_c of a steel plate. The load versus crack mouth displacement ($P - V$) record of the test gives $V_p=1\text{ mm}$ for $P_{u(max)}=31.6\text{ kN}$. Determine δ_c when $E = 210\text{ GPa}$, $\nu = 0.3$ and $\sigma_y = 800\text{ MPa}$ for steel. 10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
